

CSD19531Q5A 100 V N-Channel NexFET™ Power MOSFETs

Check for Samples: [CSD19531Q5A](#)

FEATURES

- Ultra-Low Q_g and Q_{gd}
- Low Thermal Resistance
- Avalanche Rated
- Pb-Free Terminal Plating
- RoHS Compliant
- Halogen Free
- SON 5 mm × 6 mm Plastic Package

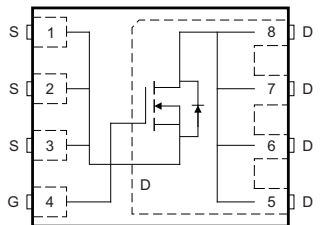
APPLICATIONS

- Primary Side Telecom
- Secondary Side Synchronous Rectifier
- Motor Control

DESCRIPTION

This 100 V, 5.3 mΩ, SON 5 mm x 6mm NexFET™ power MOSFET is designed to minimize losses in power conversion applications.

Top View



P0093-01

Product Summary

$T_A = 25^\circ\text{C}$		TYPICAL VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	100	V
Q_g	Gate Charge Total (10 V)	37	nC
Q_{gd}	Gate Charge Gate to Drain	6.6	nC
$R_{DS(on)}$	Drain-to-Source On Resistance	$V_{GS} = 6\text{ V}$	6.0
		$V_{GS} = 10\text{ V}$	5.3
$V_{GS(th)}$	Threshold Voltage	2.7	V

Ordering Information

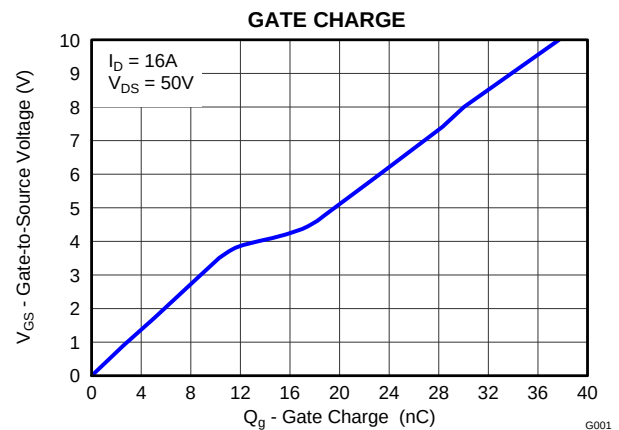
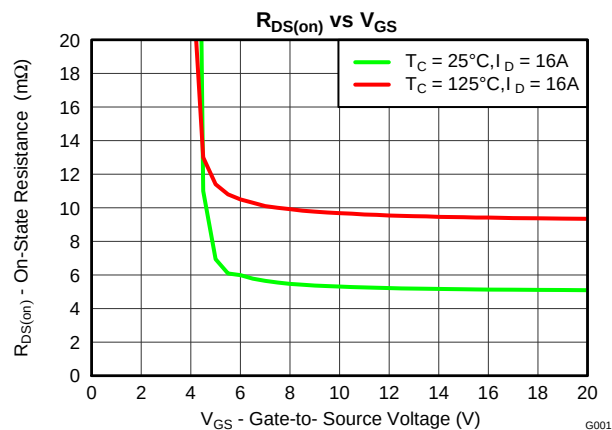
Device	Media	Qty	Package	Ship
CSD19531Q5A	13-Inch Reel	2500	SON 5 x 6 mm Plastic Package	Tape and Reel
CSD19531Q5AT	7-Inch Reel	250		

Absolute Maximum Ratings

$T_A = 25^\circ\text{C}$		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	100	V
V_{GS}	Gate-to-Source Voltage	±20	V
I_D	Continuous Drain Current (Package limited)	100	A
	Continuous Drain Current (Silicon limited), $T_C = 25^\circ\text{C}$	110	
	Continuous Drain Current ⁽¹⁾	16	
I_{DM}	Pulsed Drain Current ⁽²⁾	100	A
P_D	Power Dissipation ⁽¹⁾	3.3	W
T_J, T_{STG}	Operating Junction and Storage Temperature Range	–55 to 150	°C
E_{AS}	Avalanche Energy, single pulse $I_D = 60\text{ A}$, $L = 0.1\text{ mH}$, $R_G = 25\ \Omega$	180	mJ

(1) Typical $R_{\theta JA} = 40^\circ\text{C/W}$ on a 1-inch², 2-oz. Cu pad on a 0.06-inch thick FR4 PCB.

(2) Pulse duration ≤ 300 μs, duty cycle ≤ 1%



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These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

Electrical Characteristics

(T_A = 25°C unless otherwise stated)

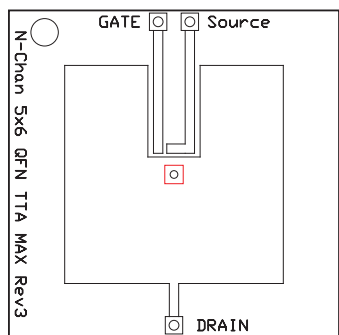
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Static Characteristics						
BV _{DSS}	Drain-to-Source Voltage	V _{GS} = 0 V, I _D = 250 μA	100			V
I _{DSS}	Drain-to-Source Leakage Current	V _{GS} = 0 V, V _{DS} = 80 V			1	μA
I _{GSS}	Gate-to-Source Leakage Current	V _{DS} = 0 V, V _{GS} = 20 V			100	nA
V _{GS(th)}	Gate-to-Source Threshold Voltage	V _{DS} = V _{GS} , I _D = 250 μA	2.2	2.7	3.3	V
R _{DS(on)}	Drain-to-Source On Resistance	V _{GS} = 6 V, I _D = 16 A		6.0	7.8	mΩ
		V _{GS} = 10 V, I _D = 16 A		5.3	6.4	mΩ
g _{fs}	Transconductance	V _{DS} = 10 V, I _D = 16 A		82		S
Dynamic Characteristics						
C _{iss}	Input Capacitance	V _{GS} = 0 V, V _{DS} = 50 V, f = 1 MHz		2980	3870	pF
C _{oss}	Output Capacitance			560	728	pF
C _{rss}	Reverse Transfer Capacitance			13.0	16.9	pF
R _G	Series Gate Resistance			1.3	2.6	Ω
Q _g	Gate Charge Total (10 V)	V _{DS} = 50 V, I _D = 16 A		37	48	nC
Q _{gd}	Gate Charge Gate to Drain			6.6		nC
Q _{gs}	Gate Charge Gate to Source			10.5		nC
Q _{g(th)}	Gate Charge at V _{th}			7.3		nC
Q _{oss}	Output Charge	V _{DS} = 50 V, V _{GS} = 0 V		97		nC
t _{d(on)}	Turn On Delay Time	V _{DS} = 50 V, V _{GS} = 10 V, I _{DS} = 16 A, R _G = 0 Ω		6.0		ns
t _r	Rise Time			5.8		ns
t _{d(off)}	Turn Off Delay Time			18.4		ns
t _f	Fall Time			5.2		ns
Diode Characteristics						
V _{SD}	Diode Forward Voltage	I _{SD} = 16 A, V _{GS} = 0 V		0.8	1	V
Q _{rr}	Reverse Recovery Charge	V _{DS} = 50 V, I _F = 16 A, di/dt = 300 A/μs		226		nC
t _{rr}	Reverse Recovery Time			148		ns

Thermal Characteristics

(T_A = 25°C unless otherwise stated)

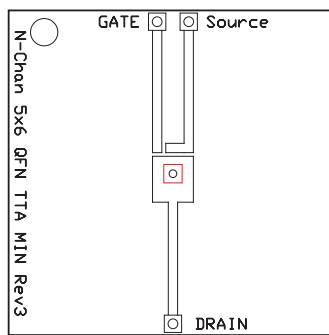
PARAMETER		MIN	TYP	MAX	UNIT
R _{θJC}	Thermal Resistance Junction to Case ⁽¹⁾			1	°C/W
R _{θJA}	Thermal Resistance Junction to Ambient ⁽¹⁾⁽²⁾			50	°C/W

- (1) R_{θJC} is determined with the device mounted on a 1-inch² (6.45-cm²), 2-oz. (0.071-mm thick) Cu pad on a 1.5-inch × 1.5-inch (3.81-cm × 3.81-cm), 0.06-inch (1.52-mm) thick FR4 PCB. R_{θJC} is specified by design, whereas R_{θJA} is determined by the user's board design.
- (2) Device mounted on FR4 material with 1-inch² (6.45-cm²), 2-oz. (0.071-mm thick) Cu.



M0137-01

Max $R_{\theta JA} = 50^{\circ}\text{C/W}$
when mounted on
1 inch² (6.45 cm²) of
2-oz. (0.071-mm thick)
Cu.



M0137-02

Max $R_{\theta JA} = 115^{\circ}\text{C/W}$
when mounted on a
minimum pad area of
2-oz. (0.071-mm thick)
Cu.

Typical MOSFET Characteristics

($T_A = 25^{\circ}\text{C}$ unless otherwise stated)

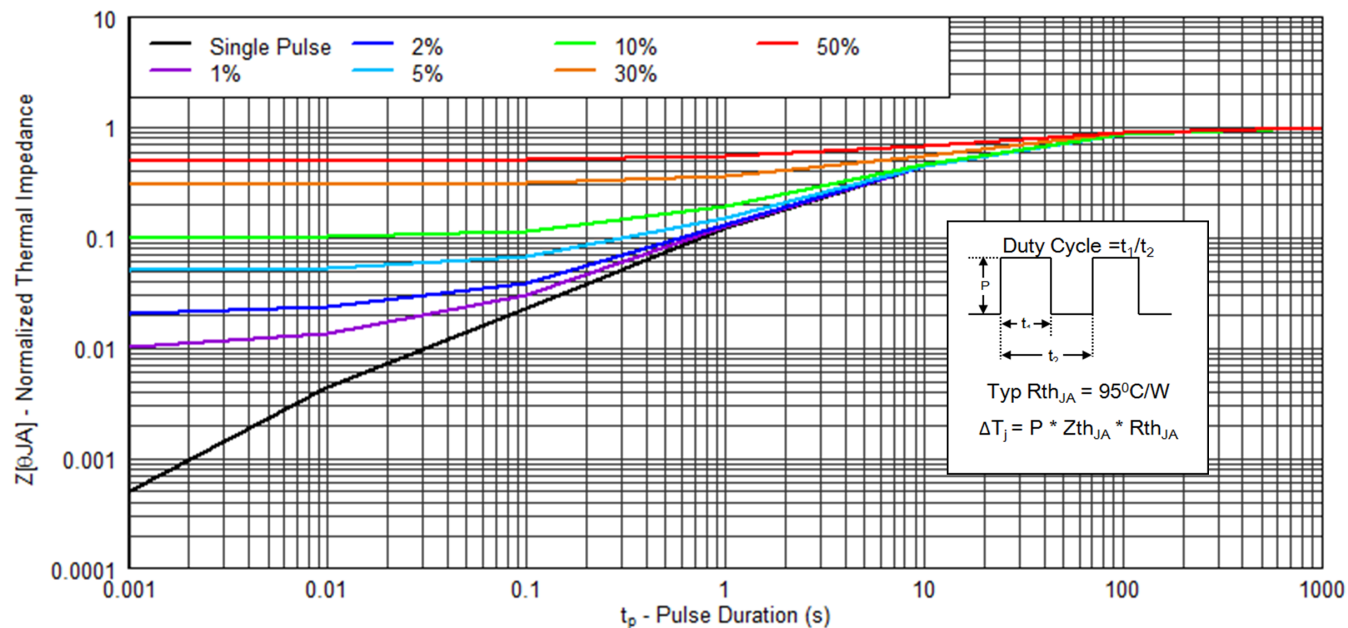


Figure 1. Transient Thermal Impedance

Typical MOSFET Characteristics (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

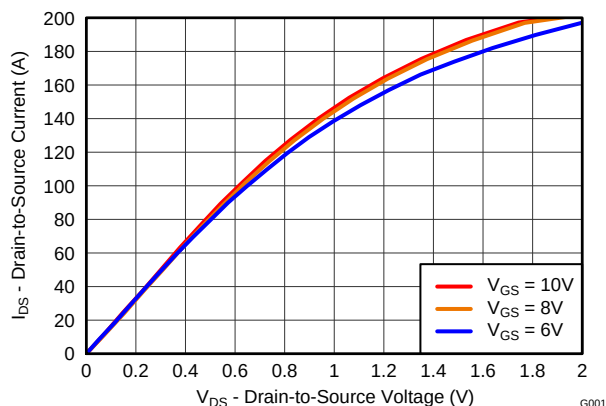


Figure 2. Saturation Characteristics

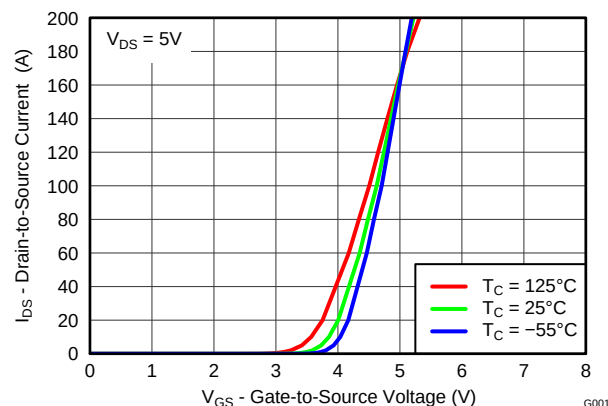


Figure 3. Transfer Characteristics

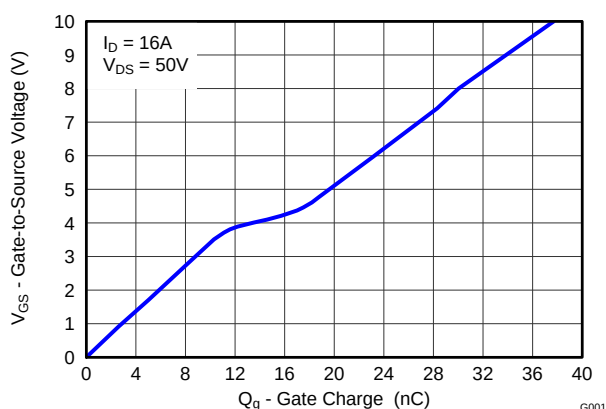


Figure 4. Gate Charge

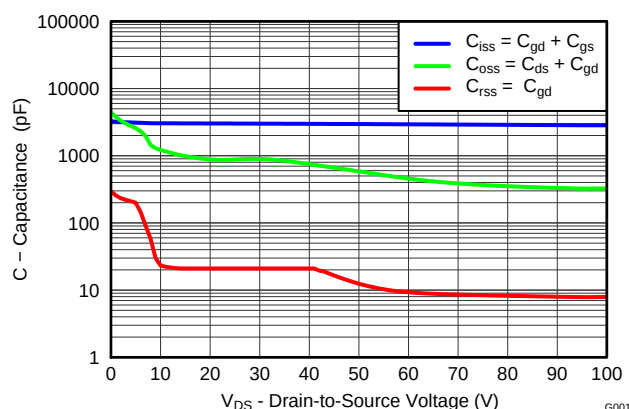


Figure 5. Capacitance

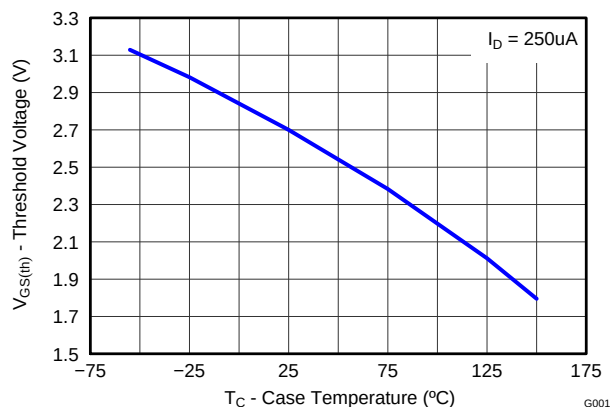


Figure 6. Threshold Voltage vs Temperature

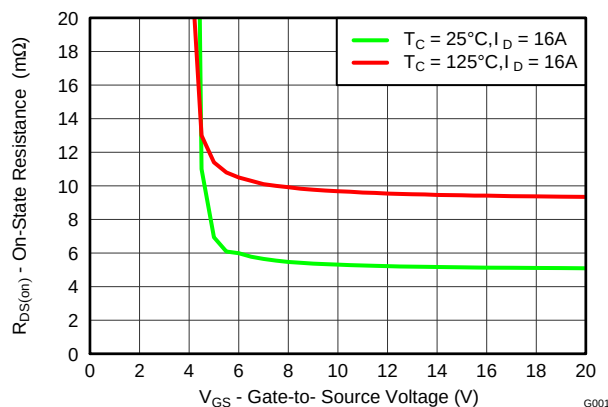


Figure 7. On-State Resistance vs Gate-to-Source Voltage

Typical MOSFET Characteristics (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

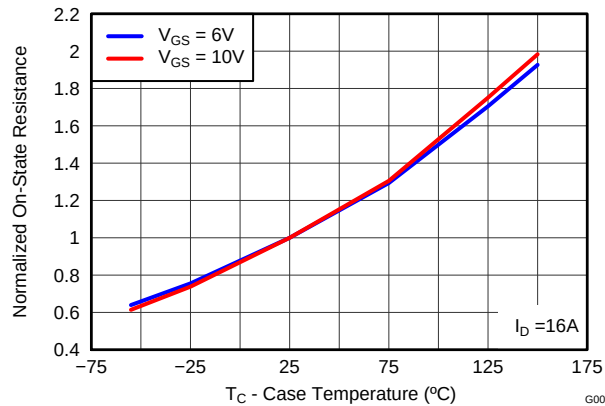


Figure 8. Normalized On-State Resistance vs Temperature

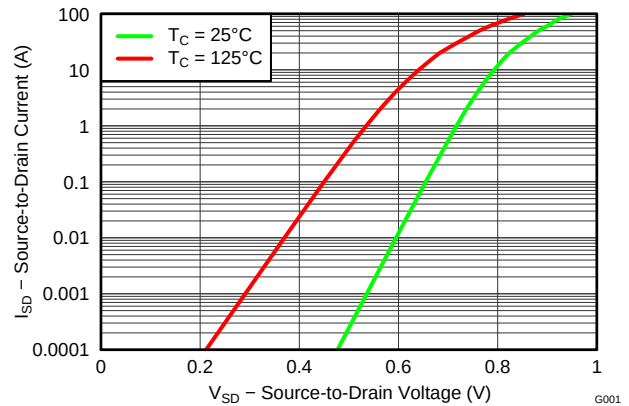


Figure 9. Typical Diode Forward Voltage

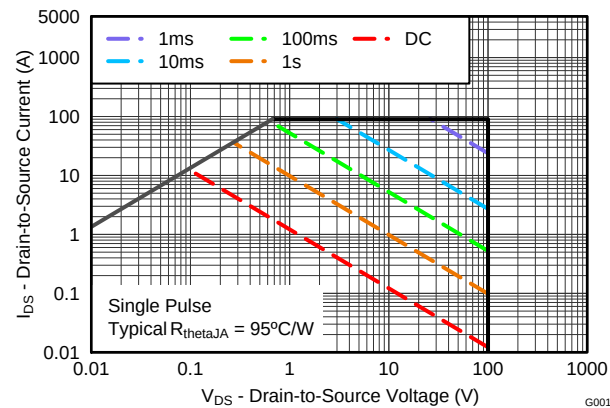


Figure 10. Maximum Safe Operating Area

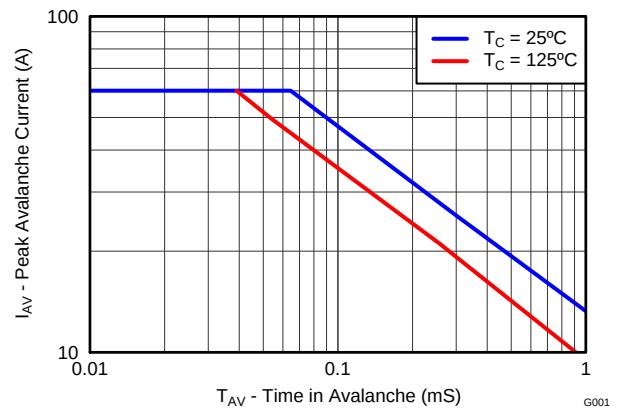


Figure 11. Single Pulse Unclamped Inductive Switching

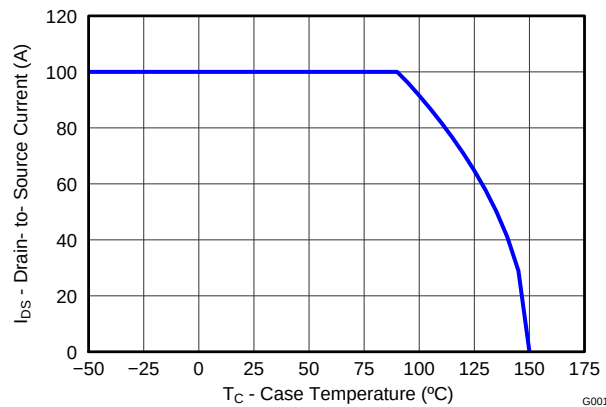
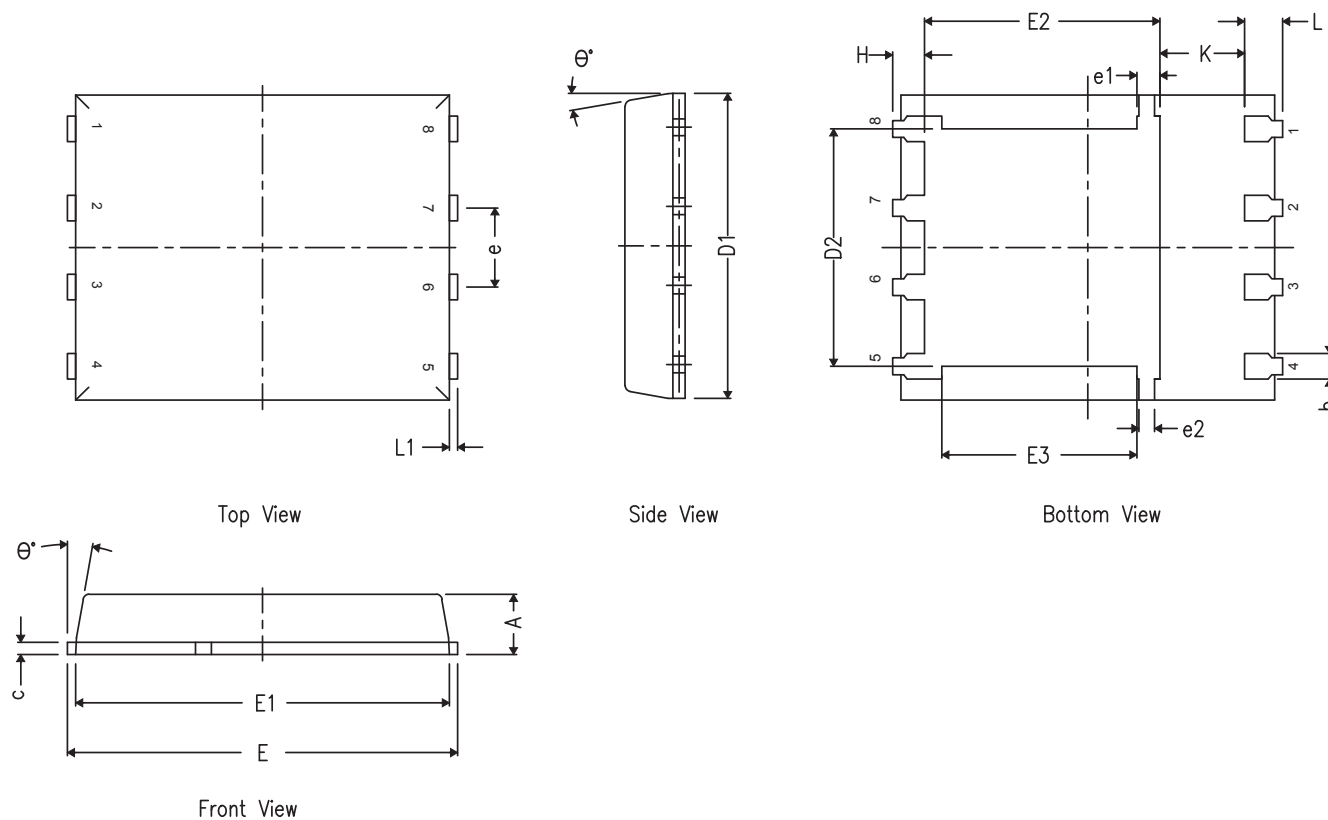


Figure 12. Maximum Drain Current vs Temperature

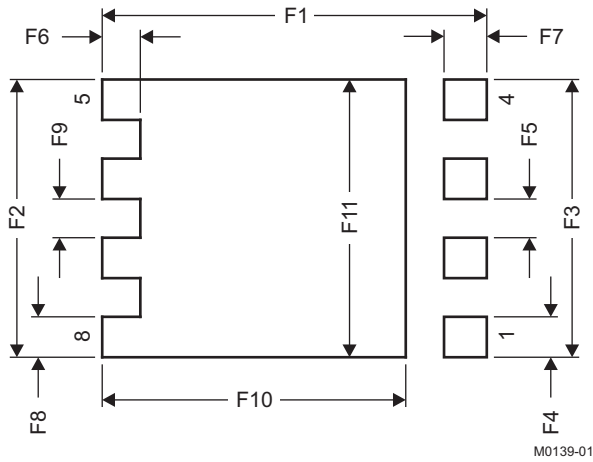
Mechanical Data

Q5A Package Dimensions



DIM	MILLIMETERS		
	MIN	NOM	MAX
A	0.90	1.00	1.10
b	0.33	0.41	0.51
c	0.20	0.25	0.34
D1	4.80	4.90	5.00
D2	3.61	3.81	4.02
E	5.90	6.00	6.10
E1	5.70	5.75	5.80
E2	3.38	3.58	3.78
E3	3.03	3.13	3.23
e	1.17	1.27	1.37
e1	0.27	0.37	0.47
e2	0.15	0.25	0.35
H	0.41	0.56	0.71
K	1.10		
L	0.51	0.61	0.71
L1	0.06	0.13	0.20
θ	0°		12°

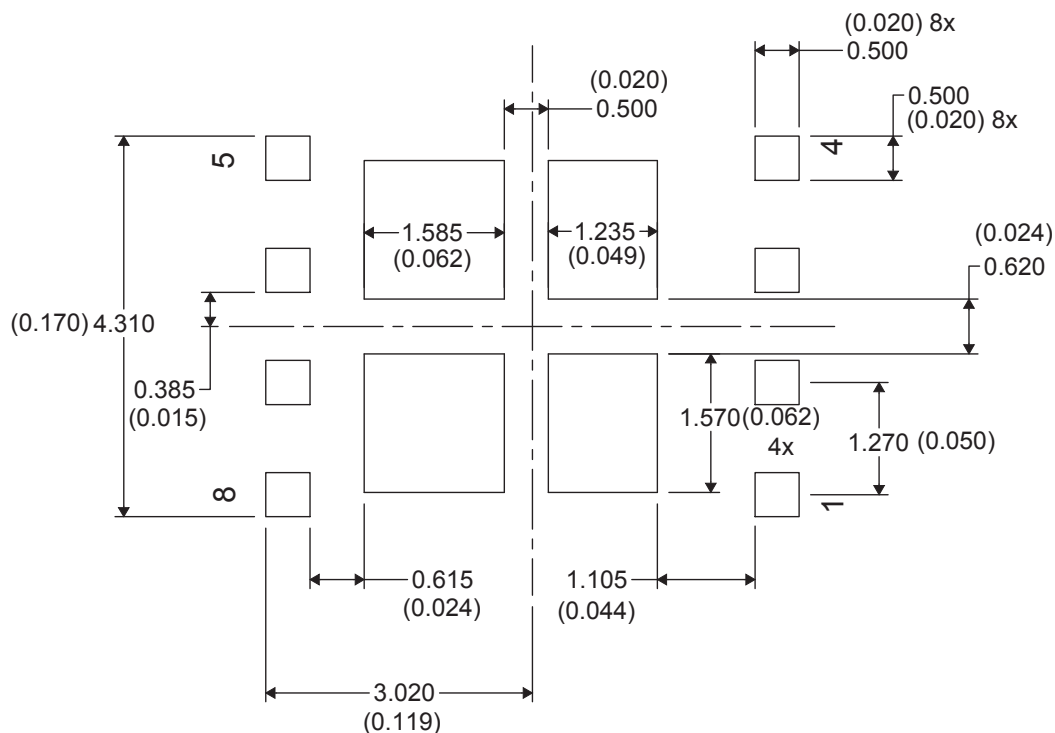
Recommended PCB Pattern



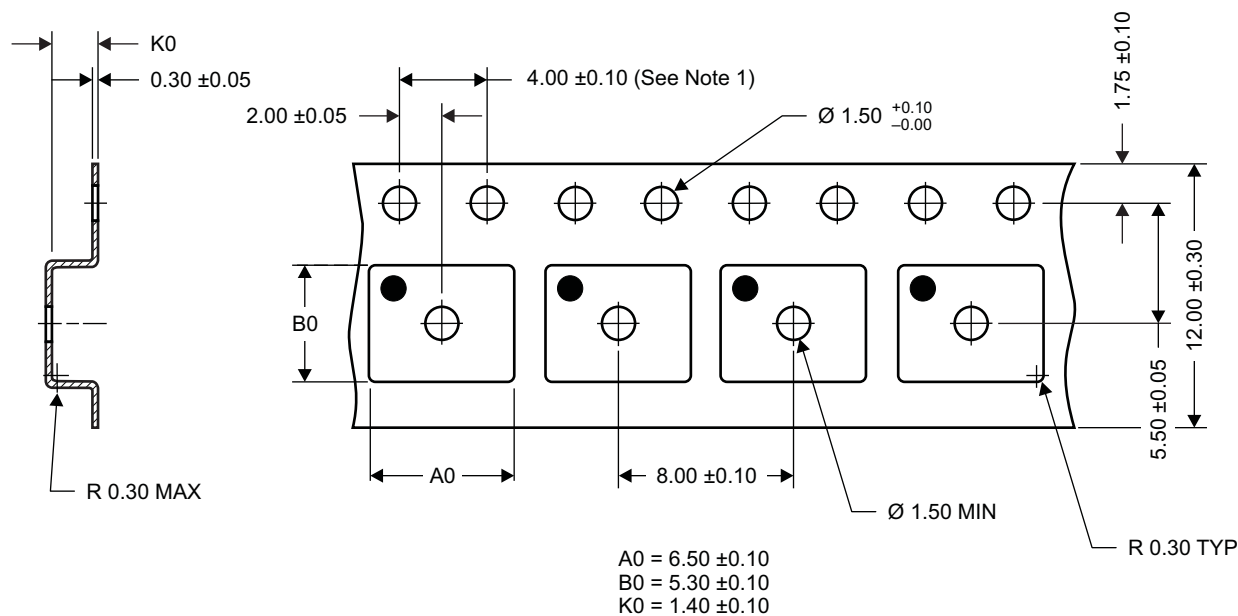
DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
F1	6.205	6.305	0.244	0.248
F2	4.46	4.56	0.176	0.18
F3	4.46	4.56	0.176	0.18
F4	0.65	0.7	0.026	0.028
F5	0.62	0.67	0.024	0.026
F6	0.63	0.68	0.025	0.027
F7	0.7	0.8	0.028	0.031
F8	0.65	0.7	0.026	0.028
F9	0.62	0.67	0.024	0.026
F10	4.9	5	0.193	0.197
F11	4.46	4.56	0.176	0.18

For recommended circuit layout for PCB designs, see application note [SLPA005](#) – *Reducing Ringing Through PCB Layout Techniques*.

Recommended Stencil Opening



Q5A Tape and Reel Information



M0138-01

Notes:

- 10-sprocket hole-pitch cumulative tolerance ±0.2
- Camber not to exceed 1 mm in 100 mm, noncumulative over 250 mm
- Material: black static-dissipative polystyrene
- All dimensions are in mm (unless otherwise specified)
- A0 and B0 measured on a plane 0.3 mm above the bottom of the pocket

REVISION HISTORY

Changes from Original (September 2013) to Revision A	Page
• Added more information to description	1
• Added small reel order number	1
• Removed $T_C = 25^{\circ}\text{C}$ condition from continuous drain current (package limited) in Absolute Maximum Ratings table	1
• Changed Typ $R_{thJA} = 99^{\circ}\text{C/W}$ to $R_{thJA} = 100^{\circ}\text{C/W}$ in Figure 1	3

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
CSD19531Q5A	ACTIVE	SON	DQJ	8	2500	Pb-Free (RoHS Exempt)	CU SN	Level-1-260C-UNLIM	-55 to 150	CSD19531	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

TBD: The Pb-Free/Green conversion plan has not been defined.

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.

Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

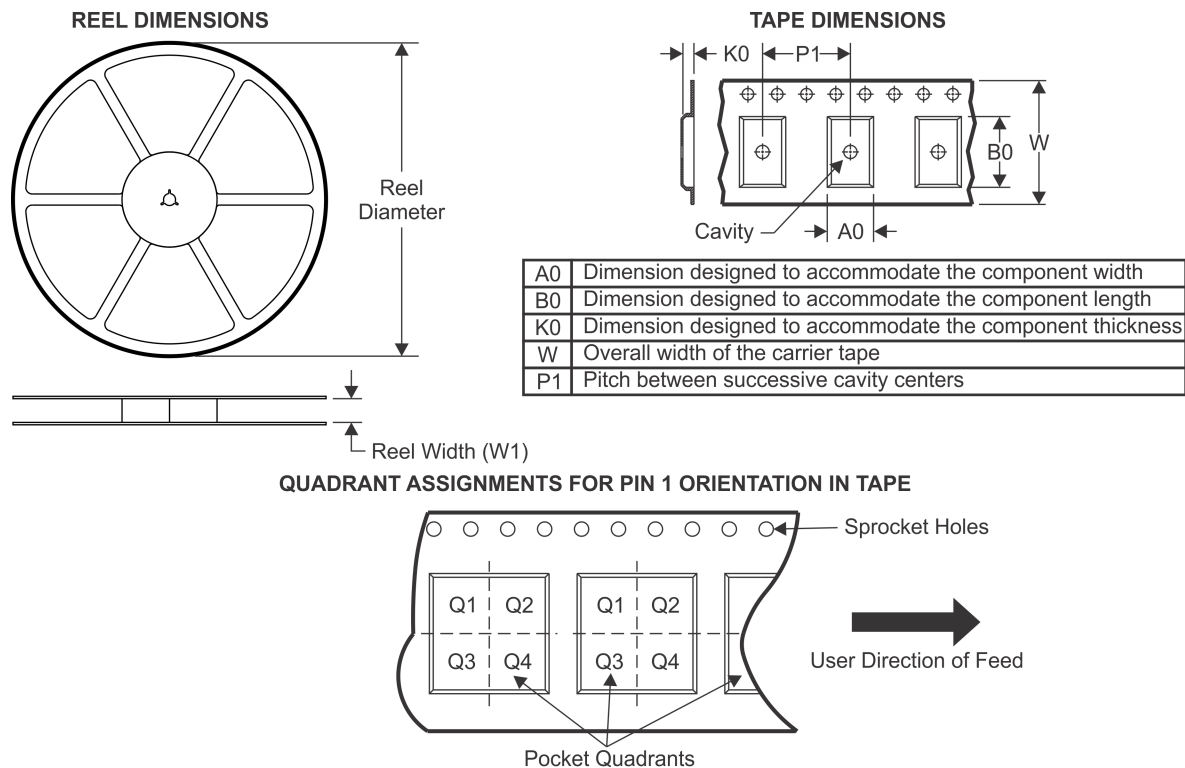
(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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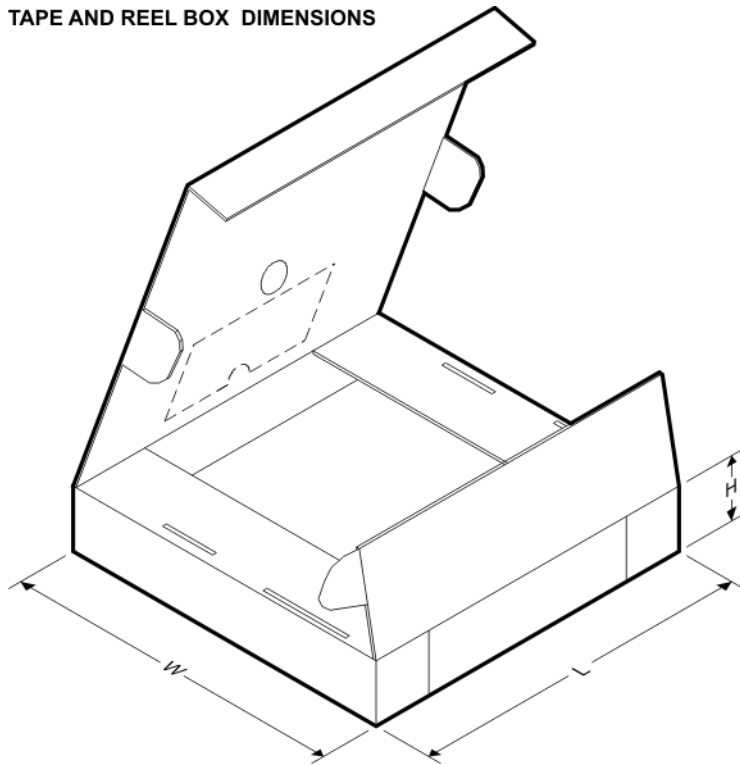
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TAPE AND REEL INFORMATION


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CSD19531Q5A	SON	DQJ	8	2500	330.0	12.4	6.3	5.3	1.2	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CSD19531Q5A	SON	DQJ	8	2500	340.0	340.0	38.0

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